A. DATASET & DATABASE

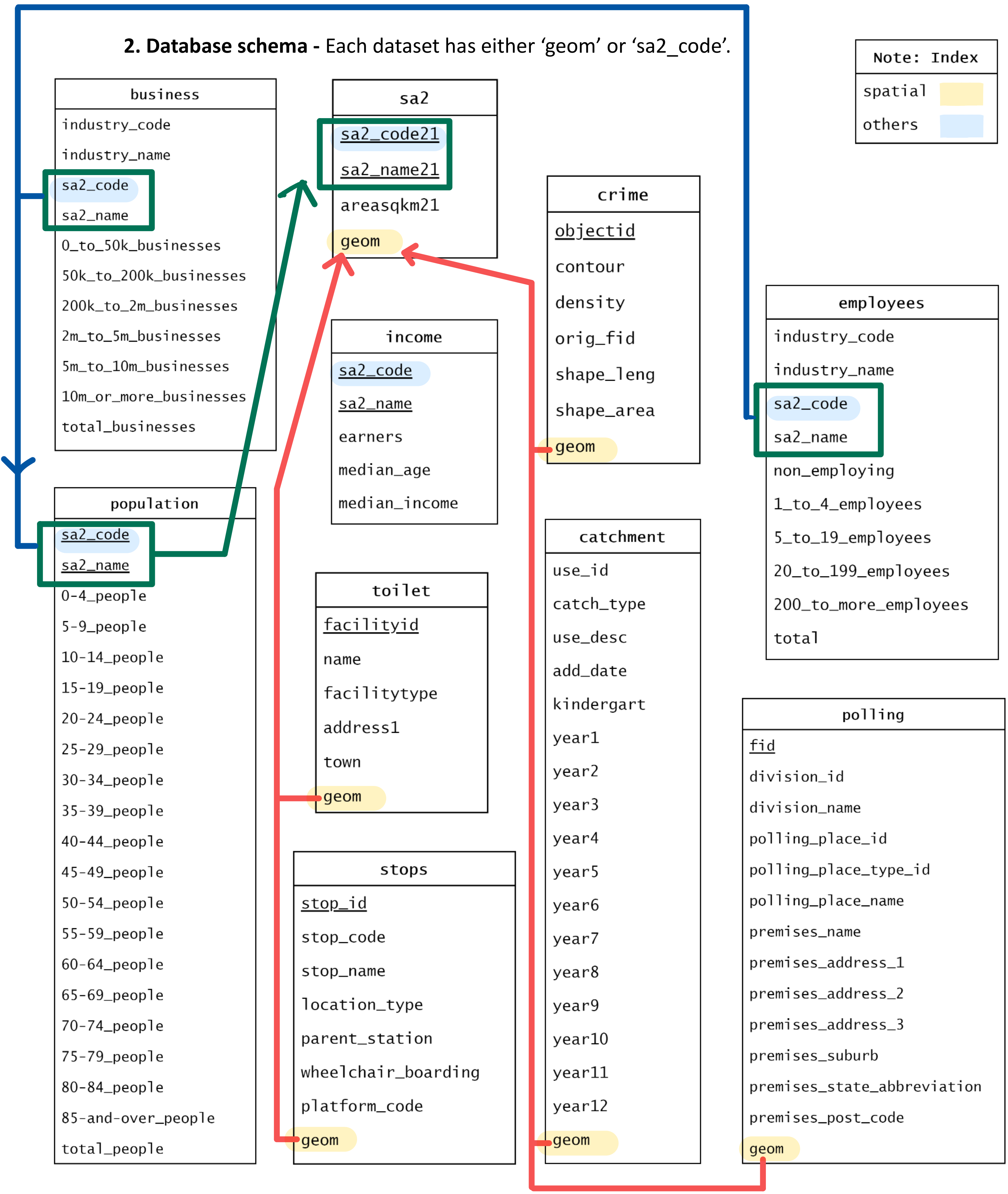
	Dataset	Type	Source	Definition
1	income	csv	Australia Bureau of Statistics	income
2	polling	csv	Australian Electoral Commission	federal election polls
3	business	csv	Australian Bureau of Statistics	turnover size
4	stops	txt	Transport for NSW	public transport stops
5	catchment_future	shp	NSW Department of Education	school
	catchment_secondary	shp	NSW Department of Education	school
	catchment_primary	shp	NSW Department of Education	school
6	sa2	shp	Australian Bureau of Statistics	statistical area level 2
7	toilet	json	Department of Health	public toilets
8	crime	shp	Bureau of Crime Statistics and Research	domestic assault
9	employees	csv	Australian Bureau of Statistics	employee size

1. Pre-process data (python)

The common process is dealing with NaNs, dropping unnecessary rows, columns, and checking data types. We always drop NaN rows in columns like **longitude, latitude** or **geometry.** For data sets with **longitude** and **latitude**, create new **geom** column from that. For spatial data sets with **geometry** column (polygon), convert to new **geom** column (multi-polygon). We also identify primary columns by comparing the total rows to the total number of unique values that each column holds. If it's equal, then it's the primary column.

1	> income	-Replace NaN ("np") in numerical columns with 0. Because NaN income has same meaning as $0. After that, we convert all these columns to type int64.
2	> polling	-Dropped 140 rows that has NaN in both **longitude**, **latitude**, **the_geom**
3	> business	-No NaN in the whole dataset and have correct data types.
4	> population	-No NaN in the whole dataset and have correct data types.
5	> stops	-No NaN in important columns: **longitude**, **latitude**
6	> catchment	_future -Convert numerical year into Y or N so it's the same as other 2 catchment
		_secondary & _primary -Drop column **PRIORITY** so it's same as catchment_future.
7	> sa2.shp	-Filter to "Greater Sydney". Drop NaN in **geometry**. Keep: **SA2_CODE21**, **SA2_NAME21**, **AREASQKM21**, **geom.** Convert **SA2_CODE21** to int so it matches **sa2_code** in other data
8	> toilet	-No NaN in columns: **longitude**, **latitude.**
9	> crime	-No NaN in the whole dataset
10	> employees	-Drop NaN ("w") row in important column: **sa2_code** and convert to int64. -In numerical columns, replace NaN with 0 and convert to type int64

2. Database schema - Each dataset has either ‘geom’ or ‘sa2_code’.



3. Indexes

- We create index either on geom or sa2_code.

- Indexes allow faster queries, it takes a lot of time especially for spatial join without indexes, for example, ST_Contains, ST_Overlaps, …

B. SCORE ANALYSIS

General formula : For each score they are divided either by area, or by people. As we only calculate scores for areas with at least 100 total people, these areas and people will be converted to NaN so its final z-score is also NaN. In our pre-check, sa2's area has no NaN as well as total people and young people. Hence, with a restriction of 100 people, NaN z score should only happen when the region has less than 100 people. As NaN is excluded in calculation of mean and sd, the score won't be affected.

The process of transferring score to z-score is the same:

$$Z = \frac{\overset{\text{Score}}{x} - \overset{\text{Mean}}{\mu}}{\underset{\text{SD}}{\sigma}}$$

Our score analysis: formula > index used in calculation > limitations > interesting findings (if has)

1. Retail score : is the number of businesses in retail trade divided by total people in that region (sa2) and multiply by 1000. To calculate score, we join business with sa2 by sa2_code, choose 'Retail Trade' then group by sa2_code. Note that sa2 is joined by the population before. There are some limitations in our score. Some areas have a population of less than 100, which we have already excluded these areas from our calculations. Otherwise, some of these scores would be large and affect the z score and visualization.

2. Health score : is the number of businesses in the healthcare sector, divided by total people in that region (sa2) and multiply by 1000. It's the same situation as Retail score, but we choose 'Healthcare and social assistance' from business and join it with sa2 by sa2 code. Another limitation with people is that although the score is per 1000 people, regions with less than 1000 people exists.

3. Stops score : is number of public transport stops per area square in km. We identified that stops are formed as points, and the area should contain points, so we use ST_Contains. There are no limitations in calculating stops scores.

4. Polling score : is number of federal election polling locations per area sq km. We identified that polls are formed as points, and the area should contain points, so we also use ST_Contains, and follow the same steps with calculating stops score.

5. School score : is the number of schools (catchments) divided by area of each sa2 region. Since school can be shared between 2 areas, we use ST_Overlaps so 1 school can be counted for all areas cutting it. We then get the school score and convert it into z score. The limitation is that raw school data contains duplicated school areas, however most of them are different in catch types so we leave it. It's interesting that future catchment differs from primary and secondary.

6. Toilet score : is the total counts of public toilets divided by area of sa2 region. We count the number of public toilets that each sa2 region area contains. Then we get the toilet rate per area and convert it into z score.

7. Crime score : is crime area divided by area of sa2 region. Raw crime data has some NaN crime areas, that means z score of these areas will also be nan. However, nan z score is only for areas with less than 100 people, hence we replace the crime area of nan with 0, which means that area has no crime. To calculate score, we found overlapping area between crime area and sa2 region area. Then we get the crime rate per area and convert it into z score.

There are some limitations. We do not know if nan crime data is due to the crime area being super small, super big or no crime because the data isn't full. However, we are giving a good score (0 crime) for these unknown areas, therefore, the final z score may be biased. Also, the crime score doesn't reflect the area's whole "crime" because the data set is limited to domestic assaults only, it doesn't include house breaking, stealing etc…

An interesting finding is there's a weak positive correlation between crime score and median income (0.22), which indicates almost no relationship between crime and income. However, we expect that higher income areas means less crime, since crime can come from financial stress (income, employment), social stress (social standing, inequality).

	crime_score	median_income
crime_score	1.000000	0.215537
median_income	0.215537	1.000000

8. Non-employing score : is the number of non employing construction industries per 1000 people, to be more specific is the number of industries which didn't hire employees. To calculate score, we join employees with sa2 by sa2 code. There are limitations, the nan exists, and the score is per 1000 people, but the regions with less than 1000 people exist.

C. OVERVIEW

Extended formula

- while all raw score (stops, schools, polling, retail and health) is all plus score

$$\text{Score} = S(z_{\text{retail}} + z_{\text{health}} + z_{\text{stops}} + z_{\text{polls}} + z_{\text{schools}})$$

- we extend the base formula with the additional data set.

- the additional data set has:

1. **Toilet score** is a plus score because it shows positive aspects of a region relates to public facilities. Emerita professor of inclusive urban planning said that "*most local planning authorities do not refer to toilet provision within their plans*". Therefore, we believe that regions with higher ratio of number of toilets/area show greater government care in public space planning, hence has a part in increasing livability index of the area.

On the other hand, 2 other datasets: crime and non-employing focus on negative impacts.

2. **Crime score**, to e more specific, domestic assault, should decrease area livability score.

3. **Non-employing score**, since construction industry in business need large amount of human resources, non-employing show the un-used human resources.

- therefore, the extended formula by additional data sets is

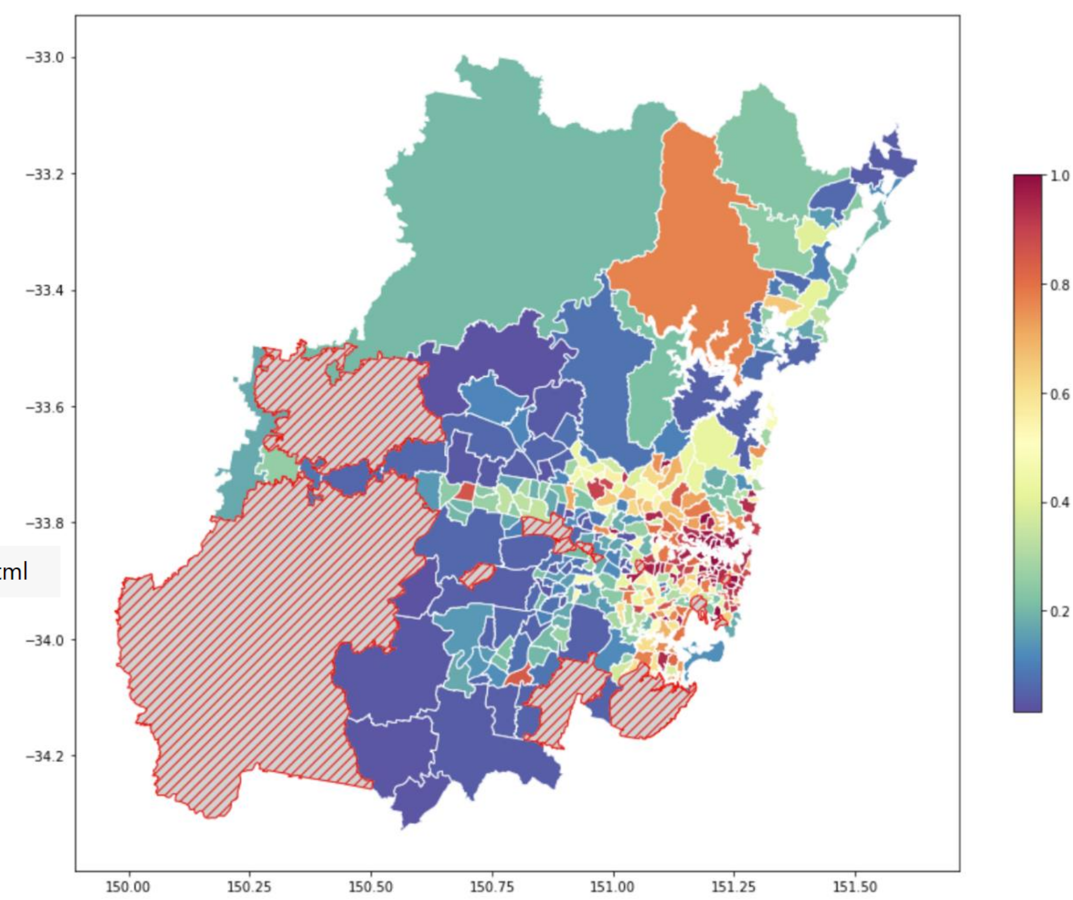
$$\text{Score} = S(Z\text{retail} + Z\text{health} + Z\text{stops} + Z\text{polls} + Z\text{schools} + Z\text{toilet} - Z\text{crime} - Z\text{employee})$$

Sigmoid score

$$S(x) = \frac{1}{1 + e^{-x}}$$

An interactive version of the map is found under the working dir

interactive_cc09_g2.html



The Sigmoid function converts the summary (+ if the score has a positive impact on well_resourced, - if not such as crime) of the z score to a probability value between 0 to 1, which represents the degree of the well-resourced score. Based on the plot and quantile, we can see areas with high well_resourced scores are concentrated in the city of Sydney.

D. CORRELATION BETWEEN WELL_RESOURCED & MEDIAN INCOME

	well_resourced	median_income
well_resourced	1.000000	0.504201
median_income	0.504201	1.000000

For a correlation coefficient of 0.504, there is a moderate positive relationship between well-resourced score & median income.

USEFULNESS This plot carries a regression line to present correlations more clearly. It can also provide hints of regional development and improvement by understanding the relationship between median income and well-resourced.

LIMITATIONS However, identifying the sa2_region represented by the point in the plot may be a little bit tricky. In addition, to get a more accurate well-resourced score, it is necessary to explore a wider range of resource data.